

Elie Gambache

Open-source library creator & systems architect — **Compose ecosystem pioneer**

I build the **invisible infrastructure** other developers build on: native desktop runtimes, cross-language platform bridges, and the primitives that make Kotlin a first-class citizen on every screen. Creator of **Nucleus**, the native desktop platform for the JVM, and 30+ open-source libraries for the Compose Multiplatform ecosystem — trusted by JetBrains, ETH Zürich, and apps with 280K+ downloads.



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github.com/kdroidFilter

nucleusframework.dev

2,938

Stars · personal + NucleusFramework
org

149

Followers

#6

GitHub committer in Israel

PROFILE

Self-directed open-source engineer specializing in Kotlin Multiplatform and Compose Desktop. I work at the layer beneath the application — wrapping raw OS APIs (Win32, AppKit, D-Bus, MediaFoundation, WinRT) in clean Kotlin, bridging Rust, C++, and Go into the JVM through JNI/FFM, and shipping signed, self-updating installers across macOS, Windows, and Linux from a single codebase. My work fills the gaps the ecosystem left open, and it composes: every library is designed to be used with the others.

FLAGSHIP WORK

Nucleus — **FRAMEWORK**

★ 304 · v2.0 · Maven Central

The native desktop platform for the JVM

A Gradle-plugin framework that turns Compose Multiplatform into a complete desktop stack: native window decorations, deep OS integration, code signing, notarization, and native installers — all from one DSL. Runs as a GraalVM native image (~0.5 s cold start, 60 MB RAM, 38 MB binary) or on the JVM with AOT cache. 30+ modules; push a tag → get signed installers in 16 packaging formats. *“Electron was the pioneer — Kotlin + Compose + Nucleus is what comes next.”*

Gradle Plugin

GraalVM · JVM

FFM · JNI

Decorated Window

Tray · Notifications · Hotkeys

Auto-Update

ComposeMediaPlayer

★ 475

The only Compose player on native codecs – no VLC, no MPV

Every other JVM media player shells out to an external process. This one talks directly to MediaFoundation (Windows), AVPlayer (macOS/iOS), GStreamer (Linux), ExoPlayer (Android), and the browser player — one @Composable, zero external dependencies, instant launch.

Supported by [JetBrains](#) · used by [Pushscroll](#) (280K+ downloads, Google Play & App Store) & AeroDL

ComposeNativeTray

★ 398

A true native system tray – not AWT, not emulated

A declarative Compose API over the OS-level system tray, each platform on its own native backend: Go + D-Bus on Linux, C++ on Windows, native APIs on macOS.

Sponsored by [ETH Zürich](#) · used by [AB Download Manager](#) (12K★)

SELECTED OPEN-SOURCE LIBRARIES

ComposeNativeWebView

★ 162

OS-native WebView (WKWebView / WebView2 / WebKitGTK) via Wry + UniFFI — no Chromium, 90% smaller binaries.

ComposePdfReader

★ 59

KMP PDF rendering & text extraction on pdfium — Compose-first API with progressive rendering & selectable text.

DesktopAccessibilityManager

[ETH Zürich](#)

Brings `announceForAccessibility()` to Compose Desktop via JNA — bridges NVDA, JAWS, VoiceOver.

AudioKt

[Rust](#)

JVM audio engine wrapping Rust's Rodio — local files, HLS, internet radio with ICY metadata.

APPLICATIONS & SYSTEMS

Zayit

4,200+ downloads

Semantic search over 7,000+ Jewish texts – fully offline

Production KMP app for reading and searching the Jewish textual library: Hebrew-aware tokenization, full-text indexing, citation-aware navigation on SQLite. Powered by Sefaria.

CORE STACK

LANGUAGES

Kotlin · Rust · C++ · Go · Python · TypeScript

FRAMEWORKS

Compose Multiplatform · KMP · React · Jetpack Compose

NATIVE BRIDGING

JNI · Java FFM · JNA · UniFFI · GraalVM native image

PLATFORM APIS

Win32 / WinRT · AppKit / AVFoundation · D-Bus / freedesktop · MediaFoundation · GStreamer

BUILD & DELIVERY

Gradle plugin dev · GitHub Actions · code signing · notarization · installers & auto-update

DOMAINS

Android MDM & enterprise · search engines & data pipelines · Hebrew NLP

RECOGNITION

◆ Open-source support from **JetBrains** (ComposeMediaPlayer)

◆ Library work **sponsored by ETH Zürich** (tray & accessibility)

◆ Libraries in production apps with **280K+ downloads**

◆ Adopted by **AB Download Manager** (12K★)

◆ **NucleusFramework** org — 818★ across 6 public repos

◆ **#6 GitHub committer in Israel** (committers.top)

ENGINEERING APPROACH

- **Infrastructure first.** Runtimes and primitives get the same care as features.
- **No leaky abstractions.** Every API is complete, honest, and hard to misuse.
- **Cross-language by design.** Right language per layer, one clean Kotlin surface.
- **Ecosystem-aware.** Libraries are built to compose, evolve, and be replaced.

FREELANCE & CONSULTING

Systems architecture

KMP library dev

Compose Desktop apps

Native integration

Gradle plugins

Android MDM

Search & data pipelines

Architecture review